



CHE 109: Engineering Chemistry – I
Summer-2025
Chemical Bonding
Chapter 3

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11.6 Classification of Solids by Type of Attraction of Units

Types of Solids

A **molecular solid** is a solid that consists of atoms or molecules held together by intermolecular forces. Many solids are of this type. Examples include solid neon, solid water (ice), and solid carbon dioxide (dry ice).

A **metallic solid** is a solid that consists of positive cores of atoms held together by a surrounding “sea” of electrons (metallic bonding). In this kind of bonding, positively charged atomic cores are surrounded by delocalized electrons. Examples include iron, copper, and silver.

An **ionic solid** is a solid that consists of cations and anions held together by the electrical attraction of opposite charges (ionic bonds). Examples are cesium chloride, sodium chloride, and zinc sulfide (but ZnS has considerable covalent character).

A **covalent network solid** is a solid that consists of atoms held together in large networks or chains by covalent bonds. Diamond is an example of a three-dimensional network solid. Every carbon atom in diamond is covalently bonded to four others, so an entire crystal might be considered an immense molecule. Graphite is another example of a network solid. It consists of carbon atoms held together by covalent bonds to form two-dimensional networks or sheets. The sheets are stacked one on top of the other, and they are held together by weak van der Waals forces between sheets. Figure 11.28 shows the structures of diamond and graphite. Table 11.4 summarizes these four types of solids.

Table 11.4 Types of Solids

Type of Solid	Structural Units	Attractive Forces Between Structural Units	Examples
Molecular	Atoms or molecules	Intermolecular forces	Ne, H ₂ O, CO ₂
Metallic	Atoms (positive cores surrounded by electron “sea”)	Metallic bonding (extreme delocalized bonding)	Fe, Cu, Ag
Ionic	Ions	Ionic bonding	CsCl, NaCl, ZnS
Covalent network	Atoms	Covalent bonding	Diamond, graphite, graphene

► Ionic Bonds

The first explanation of chemical bonding was suggested by the properties of *salts*, substances now known to be ionic. Salts are generally crystalline solids that melt at high temperatures. Sodium chloride, for example, melts at 801°C. A molten salt (the liquid after melting) conducts an electric current. A salt dissolved in water gives a solution that also conducts an electric current. The electrical conductivity of the molten salt and the salt solution results from the motion of ions in the liquids. This suggests the possibility that ions exist in certain solids, held together by the attraction of opposite charges.

9.1 Describing Ionic Bonds

An **ionic bond** is a chemical bond formed by the electrostatic attraction between positive and negative ions. The bond forms between two atoms when one or more electrons are transferred from the valence shell of one atom to the valence shell of the other. The atom that loses electrons becomes a *cation* (positive ion), and the atom that gains electrons becomes an *anion* (negative ion). ◀ Any given ion tends to attract as many neighboring ions of opposite charge as possible. When large numbers of ions gather together, they form an ionic solid. The solid normally has a regular, crystalline structure that allows for the maximum attraction of ions, given their particular sizes.

To understand why ionic bonding occurs, consider the transfer of a valence electron from a sodium atom (electron configuration $[\text{Ne}]3s^1$) to the valence shell of a chlorine atom ($[\text{Ne}]3s^23p^5$). You can represent the electron transfer by the following equation:



As a result of the electron transfer, ions are formed, each of which has a noble-gas configuration. The sodium atom has lost its 3s electron and has taken on the neon

The terms ion, anion, and cation were coined by the chemist and physicist Michael Faraday (1791–1867) during his study of the electrical properties of salt solutions.

Table 9.1 Lewis Electron-Dot Symbols for Atoms of the Second and Third Periods

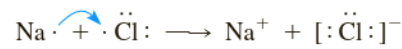
Period	1A ns^1	2A ns^2	3A ns^2np^1	4A ns^2np^2	5A ns^2np^3	6A ns^2np^4	7A ns^2np^5	8A ns^2np^6
Second	Li·	·Be·	·B·	·C·	·N·	·O·	·F·	·Ne·
Third	Na·	·Mg·	·Al·	·Si·	·P·	·S·	·Cl·	·Ar·

configuration, [Ne]. The chlorine atom has accepted the electron into its $3p$ subshell and has taken on the argon configuration, [Ne] $3s^23p^6$. Such noble-gas configurations and the corresponding ions are particularly stable. This stability of the ions accounts in part for the formation of the ionic solid NaCl. Once a cation or anion forms, it attracts ions of opposite charge. Within the sodium chloride crystal, NaCl, every Na^+ ion is surrounded by six Cl^- ions, and every Cl^- ion by six Na^+ ions. (Figure 2.19 shows the arrangement of ions in the NaCl crystal.)

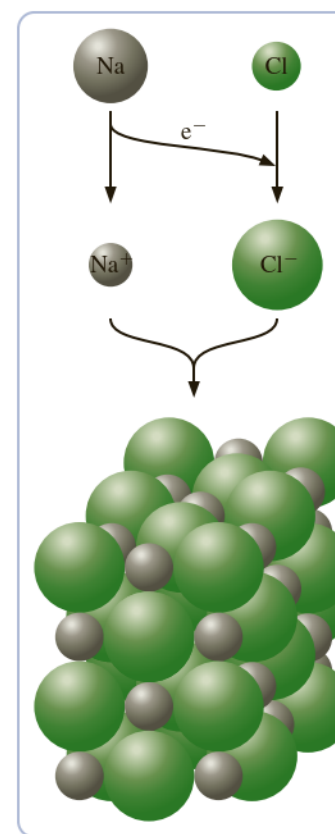
Lewis Electron-Dot Symbols

You can simplify the preceding equation for the electron transfer between Na and Cl by writing Lewis electron-dot symbols for the atoms and monatomic ions. A **Lewis electron-dot symbol** is a symbol in which the electrons in the valence shell of an atom or ion are represented by dots placed around the letter symbol of the element. Table 9.1 lists Lewis symbols and corresponding valence-shell electron configurations for the atoms of the second and third periods. Note that dots are placed one to each side of a letter symbol until all four sides are occupied. Then the dots are written two to a side until all valence electrons are accounted for. The exact placement of the single dots is immaterial. For example, the single dot in the Lewis symbol for chlorine can be written on any one of the four sides. [This pairing of dots does not always correspond to the pairing of electrons in the ground state. Thus, you write $\cdot\text{B}\cdot$ for boron, rather than $:\text{B}\cdot$, which more closely corresponds to the ground-state configuration [He] $2s^22p^1$. The first symbol better reflects boron's chemistry, in which each single electron (single dot) tends to be involved in bond formation.]

The equation representing the transfer of an electron from the sodium atom to the chlorine atom is



The noble-gas configurations of the ions are apparent from the symbols. No dots are shown for the cation. (All valence electrons have been removed, leaving the noble-gas core.) Eight dots are shown in brackets for the anion (noble-gas configuration ns^2np^6).



Example 9.1 Using Lewis Symbols to Represent Ionic Bond Formation

Gaining Mastery Toolbox

Critical Concept 9.1

To represent the formation of the ions of an ionic compound from the atoms using Lewis symbols, you show dots being transferred from the metal to nonmetal atoms to give the ions with noble-gas configurations.

Solution Essentials:

- Lewis electron-dot symbols
- Noble-gas configuration
- Formula of a binary ionic compound

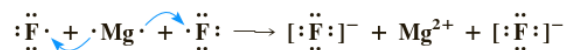
Use Lewis electron-dot symbols to represent the transfer of electrons from magnesium to fluorine atoms to form ions with noble-gas configurations.

Problem Strategy Write down the Lewis symbols for the two atoms. Note how many electrons the metal atom should lose to assume a noble-gas configuration (all of the dots, representing valence electrons) and how many electrons the nonmetal atom should gain to assume such a configuration (enough to give eight dots about the atomic symbol). Represent the transfer of electrons between ions diagrammatically in the form of an equation.

(continued)

Example 9.1 (continued)

Solution The Lewis symbols for the atoms are $\cdot\ddot{\text{F}}\cdot$ and $\cdot\text{Mg}\cdot$ (see Table 9.1). The magnesium atom loses two electrons to assume a noble-gas configuration. But because a fluorine atom can accept only one electron to fill its valence shell, two fluorine atoms must take part in the electron transfer. We can represent this electron transfer as follows:



Answer Check Check that you have the correct symbols for the atoms and that the symbols for the ions do have noble-gas configurations.

Exercise 9.1 Represent the transfer of electrons from magnesium to oxygen atoms to assume noble-gas configurations. Use Lewis electron-dot symbols.

See Problems 9.37 and 9.38.

Energy Involved in Ionic Bonding

You have seen in a qualitative way why a sodium atom and a chlorine atom might be expected to form an ionic bond. It is instructive, however, to look at the energy changes involved in ionic bond formation. From this analysis, you can gain further understanding of why certain atoms bond ionically and others do not.

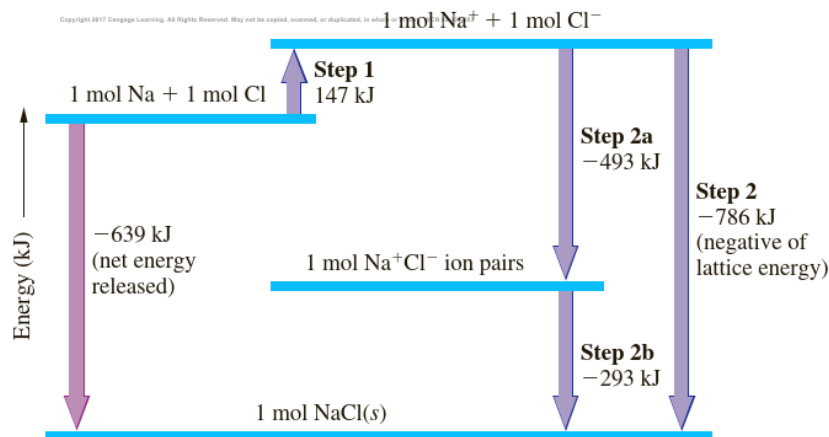
If atoms come together and bond, there should be a net decrease in energy, because the bonded state should be more stable and therefore at a lower energy level. Consider again the formation of an ionic bond between a sodium atom and a chlorine atom. You can think of this as occurring in two steps: (1) An electron is transferred between the two separate atoms to give ions. (2) The ions then attract one another to form an ionic bond. In reality, the transfer of the electron and the formation of an ionic bond occur simultaneously, rather than in discrete steps, as the atoms approach one another. But the *net* quantity of energy involved is the same whether the steps occur one after the other or at the same time.

The first step requires removal of the $3s$ electron from the sodium atom and the addition of this electron to the valence shell of the chlorine atom. Removing the electron from the sodium atom requires energy (the first ionization energy of the sodium atom, which equals 496 kJ/mol). Adding the electron to the chlorine atom releases energy (equal to -349 kJ/mol , which is the negative of the electron affinity of the chlorine atom). ◀ The overall energy of this step is $(496 - 349) \text{ kJ/mol}$, or 147 kJ/mol (Figure 9.2, Step 1). So, the process requires more energy to remove an electron from

Ionization energies and electron affinities of atoms were discussed in Section 8.6.

Figure 9.2 ▶

Energetics of ionic bonding The transfer of an electron from a Na atom to a Cl atom is not in itself energetically favorable; it requires 147 kJ/mol of energy (Step 1). However, 493 kJ of energy is released when these oppositely charged ions come together to form ion pairs (Step 2a). Moreover, additional energy (293 kJ) is released when these ion pairs form the solid NaCl crystal (Step 2b). The lattice energy released when one mole each of Na^+ and Cl^- ions react to produce $\text{NaCl}(s)$ is 786 kJ/mol , and the overall process of NaCl formation is energetically favorable, releasing 639 kJ/mol if one starts with gaseous Na and Cl atoms.



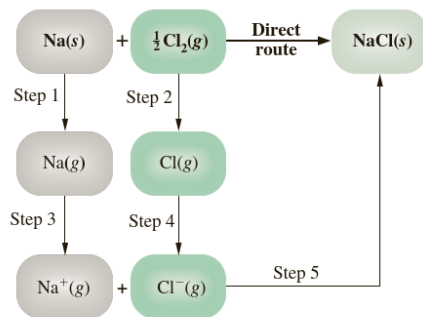


Figure 9.3 ▲

Born-Haber cycle for NaCl The formation of NaCl(s) from the elements is accomplished by two different routes. The direct route is the formation reaction (shown in boldface), and the enthalpy change is ΔH_f° . The indirect route occurs in five steps.

Lattice Energies from the Born-Haber Cycle

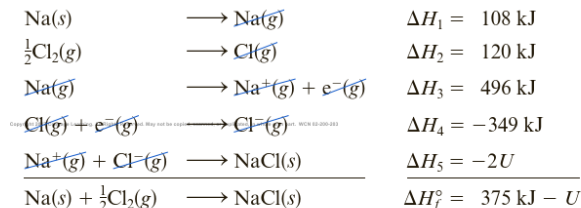
The preceding energy analysis requires us to know the lattice energy of solid sodium chloride. Direct experimental determination of the lattice energy of an ionic solid is difficult. However, this quantity can be indirectly determined from experiment by means of a thermochemical “cycle” originated by Max Born and Fritz Haber in 1919 and now called the *Born-Haber cycle*. The reasoning is based on Hess’s law.

To obtain the lattice energy of NaCl, you think of solid sodium chloride being formed from the elements by two different routes, as shown in Figure 9.3. In one route, NaCl(s) is formed directly from the elements, Na(s) and $\frac{1}{2}\text{Cl}_2(\text{g})$. The enthalpy change for this is ΔH_f° , which is given in Table 6.2 as -411 kJ per mole of NaCl. The second route consists of the following five steps, along with the enthalpy change for each. (To be precise, the ionization energy

and electron affinity are energy changes, ΔE , and we should add small corrections to give the enthalpy changes, ΔH .)

1. *Sublimation of sodium.* Metallic sodium is vaporized to a gas of sodium atoms. (*Sublimation* is the transformation of a solid to a gas.) The enthalpy change for this process, measured experimentally, is 108 kJ per mole of sodium.
2. *Dissociation of chlorine.* Chlorine molecules are dissociated to atoms. The enthalpy change for this equals the Cl—Cl bond dissociation energy, which is 240 kJ per mole of bonds, or 120 kJ per mole of Cl atoms.
3. *Ionization of sodium.* Sodium atoms are ionized to Na^+ ions. The enthalpy change is essentially the ionization energy of atomic sodium, which equals 496 kJ per mole of Na.
4. *Formation of chloride ion.* The electrons from the ionization of sodium atoms are transferred to chlorine atoms. The enthalpy change for this is the negative of the electron affinity of atomic chlorine, equal to -349 kJ per mole of Cl atoms.
5. *Formation of NaCl(s) from ions.* The ions Na^+ and Cl^- formed in Steps 3 and 4 combine to give solid sodium chloride. Because this process is just the reverse of the one corresponding to the lattice energy (breaking the solid into ions), the enthalpy change is the negative of the lattice energy. If we let U be the lattice energy, the enthalpy change for Step 5 is $-U$.

Let us write these five steps and add them. We also add the corresponding enthalpy changes, following Hess’s law.



In summing the equations, we have canceled terms that appear on both the left and right sides of the arrows. The final equation is simply the formation reaction for NaCl(s). Adding the enthalpy changes, we find that the enthalpy change for this formation reaction is $375 \text{ kJ} - U$. But the enthalpy of formation has been determined calorimetrically and equals -411 kJ. Equating these two values, we get

$$375 \text{ kJ} - U = -411 \text{ kJ}$$

Solving for U yields the lattice energy of NaCl:

$$U = (375 + 411) \text{ kJ} = 786 \text{ kJ}$$

Sodium Chloride—A Common and Important Ionic Compound

We are all familiar with sodium chloride as table salt. It is a typical ionic compound, a brittle solid with a high melting point (801°C) that conducts electricity in the molten state and in aqueous solution. The structure of solid NaCl is shown in Figure 2.13.

One source of sodium chloride is rock salt, which is found in subterranean deposits often hundreds of meters thick. It is also obtained from seawater or brine (a concentrated NaCl solution) by solar evaporation. Sodium chloride also occurs in nature as the mineral *halite*.

Sodium chloride is used more often than any other material in the manufacture of inorganic chemicals. World consumption of this substance is about 200 million tons per year. The major use of sodium chloride is in the production of other essential inorganic chemicals such as chlorine gas, sodium hydroxide, sodium metal, hydrogen gas, and sodium carbonate. It is also used to melt ice and snow on highways and roads. However, because sodium chloride is harmful to plant life and promotes corrosion of cars, its use for this purpose is of considerable environmental concern.



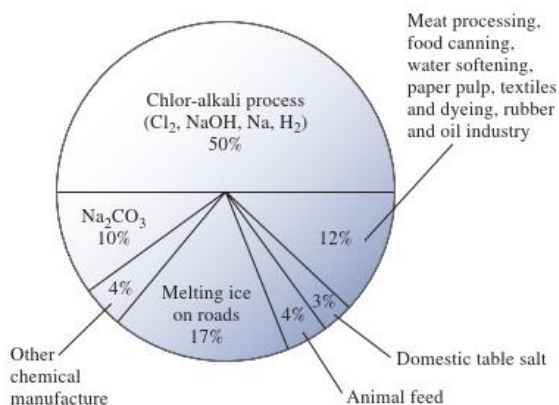
Solar evaporation process for obtaining sodium chloride.

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Underground rock salt mining.

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Uses of sodium chloride.

► Covalent Bonds

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In the preceding sections we looked at ionic substances, which are typically high-melting solids. Many substances, however, are molecular—gases, liquids, or low-melting solids consisting of molecules (Figure 9.9). A molecule is a group of atoms, frequently nonmetal atoms, strongly linked by chemical bonds. Often the forces that hold atoms together in a molecular substance cannot be understood on the basis of the attraction of oppositely charged ions (the ionic model). An obvious example is the molecule H_2 , in which the two H atoms are held together tightly and no ions are present. In 1916, Gilbert Newton Lewis proposed that the strong attractive force between two atoms in a molecule results from a **covalent bond**, *a chemical bond formed by the sharing of a pair of electrons between atoms*. ► In 1926, Walter Heitler and Fritz London showed that the covalent bond in H_2 could be quantitatively explained by the newly discovered theory of quantum mechanics. We will discuss the descriptive aspects of covalent bonding in the following sections.

G. N. Lewis (1875–1946) was professor of chemistry at the University of California at Berkeley. Well known for work on chemical bonding, he was also noted for his research in molecular spectroscopy and thermodynamics (the study of heat involved in chemical and physical processes).

Figure 9.9 ▶
Two molecular substances

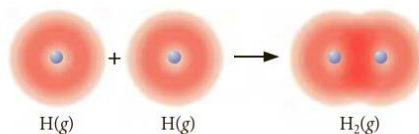
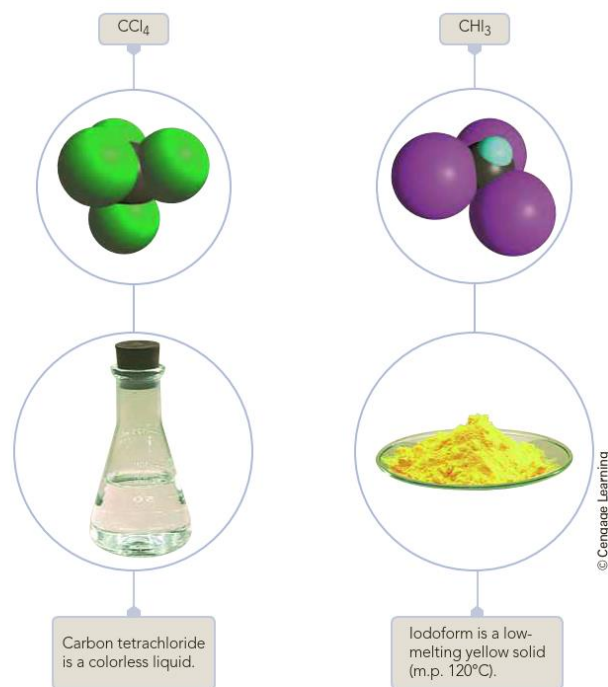


Figure 9.10 ▲
The electron probability distribution for the H_2 molecule
The electron density (shown in red) occupies the space around both atoms.

9.4 Describing Covalent Bonds

Consider the formation of a covalent bond between two H atoms to give the H_2 molecule. As the atoms approach one another, their $1s$ orbitals begin to overlap. Each electron can then occupy the space around both atoms. In other words, the two electrons can be shared by the atoms (Figure 9.10). The electrons are attracted simultaneously by the positive charges of the two hydrogen nuclei. This attraction that bonds the electrons to both nuclei is the force holding the atoms together. Although ions do not exist in H_2 , the force that holds the atoms together can still be regarded as arising from the attraction of oppositely charged particles: nuclei and electrons.

It is interesting to see how the potential energy of the atoms changes as they approach and then bond. Figure 9.11 shows the potential energy of the atoms for various distances between nuclei. The potential energy of the atoms when they are some distance apart is indicated by a position on the potential-energy curve at the far right. As the atoms approach (moving from right to left on the potential-energy curve), the potential energy gets lower and lower. The decrease in energy is a reflection of the bonding of the atoms. Eventually, as the atoms get close enough, the repulsion of the positive charges on the nuclei becomes larger than the attraction of electrons for nuclei. In other words, the potential energy reaches a minimum value and then increases. The distance between nuclei at this minimum energy is called the *bond length* of H_2 . It is the normal distance between nuclei in the molecule.

Now imagine the reverse process. You start with the H_2 molecule, the atoms at their normal bond length apart (at the minimum of the potential-energy curve). To separate the atoms in the molecule, energy must be added (you move along the curve to the flat portion at the right). The energy that must be added is called the *bond dissociation energy*. The larger the bond dissociation energy, the stronger the bond.

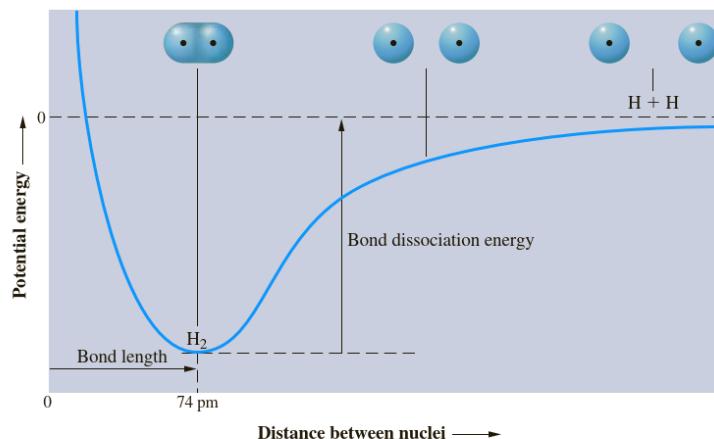


Figure 9.11 ◀
Potential-energy curve for the formation of a stable molecule occurs at a distance corresponding to a minimum in the potential-energy curve.

Lewis Formulas

You can represent the formation of the covalent bond in H_2 from atoms as follows:



This uses the Lewis electron-dot symbol for the hydrogen atoms and represents the covalent bond by a pair of dots. Recall that the two electrons from the covalent bond spend part of the time in the region of each atom. In this sense, each atom in H_2 has a helium configuration. We can draw a circle about each atom to emphasize this.



The formation of a bond between H and Cl to give an HCl molecule can be represented in a similar way.



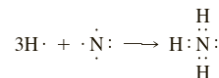
As the two atoms approach each other, unpaired electrons on each atom pair up to form a covalent bond. The pair of electrons is shared by the two atoms. Each atom then acquires a noble-gas configuration of electrons, the H atom having two electrons about it (as in He), and the Cl atom having eight valence electrons about it (as in Ar).

A formula using dots to represent valence electrons is called a **Lewis electron-dot formula**. An electron pair represented by a pair of dots in such a formula is either a **bonding pair** (an electron pair shared between two atoms) or a **lone**, or **nonbonding, pair** (an electron pair that remains on one atom and is not shared). For example,



Bonding pairs are often represented by dashes rather than by pairs of dots.

Frequently, the number of covalent bonds formed by an atom equals the number of unpaired electrons shown in its Lewis symbol. Consider the formation of NH_3 .

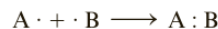


The numbers of unpaired electrons in the Lewis symbols for the atoms of elements in Groups 1A and 3A equal the group numbers. But except for the first elements of these groups, the atoms usually form ionic bonds.

Each bond is formed between an unpaired electron on one atom and an unpaired electron on another atom. In many instances, the number of bonds formed by an atom in Groups 4A to 7A equals the number of unpaired electrons, which is eight minus the group number. For example, a nitrogen atom (Group 5A) forms $8 - 5 = 3$ covalent bonds. ◀

Coordinate Covalent Bonds

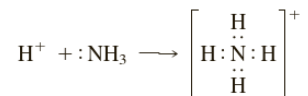
When bonds form between atoms that both donate an electron, you have



However, it is possible for both electrons to come from the same atom. A **coordinate covalent bond** is a bond formed when both electrons of the bond are donated by one atom:



A coordinate covalent bond is not essentially different from other covalent bonds; it involves the sharing of a pair of electrons between two atoms. An example is the formation of the ammonium ion, in which an electron pair on the N atom of NH_3 forms a bond with H^+ .



The new N—H bond is clearly identical to the other N—H bonds.

Octet Rule

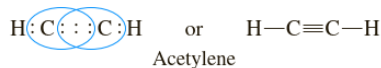
In each of the molecules we have described so far, the atoms have obtained noble-gas configurations through the sharing of electrons. Atoms other than hydrogen have obtained eight electrons in their valence shells; hydrogen atoms have obtained two. *The tendency of atoms in molecules to have eight electrons in their valence shells (two for hydrogen atoms)* is known as the **octet rule**. Many of the molecules we will discuss follow the octet rule. Some do not.

Multiple Bonds

In the molecules we have described so far, each of the bonds has been a **single bond**—that is, a covalent bond in which a single pair of electrons is shared by two atoms. But it is possible for atoms to share two or more electron pairs. A **double bond** is a covalent bond in which two pairs of electrons are shared by two atoms. A **triple bond** is a covalent bond in which three pairs of electrons are shared by two atoms. As examples, consider ethylene, C_2H_4 , and acetylene, C_2H_2 . Their Lewis formulas are

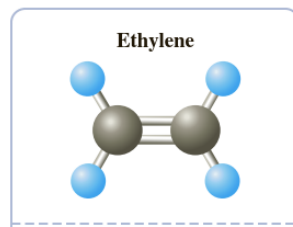


Ethylene



Acetylene

Note the octet of electrons on each C atom. Double bonds form primarily with C, N, O, and S atoms. Triple bonds form mostly to C and N atoms.





Daily Life

A **CHEMIST** Looks at ...

Chemical Bonds in Nitroglycerin

Nitroglycerin gained a nasty reputation soon after its discovery in 1846. Unless kept cold, the pale yellow, oily liquid detonates from even the mildest vibration. In the French film *The Wages of Fear*, four men agree to drive two trucks loaded with nitroglycerin over mountainous roads from a remote village in South America. They are to drive the nitroglycerin to an out-of-control oil-well fire, where the explosive will be used to close off the well. On the way, they find the road obstructed by a huge boulder that has rolled across it, and they decide to blast the boulder out of their way using some of their explosive cargo. You see one of the men gingerly pouring nitroglycerin from a thermos bottle down a stick and into a hole in the rock. Beads of sweat form on his contorted face, while he tries desperately to suspend his breathing and any extra movement that might set off the nitroglycerin.

Here is the structure of nitroglycerin (also see the molecular model in Figure 9.12):

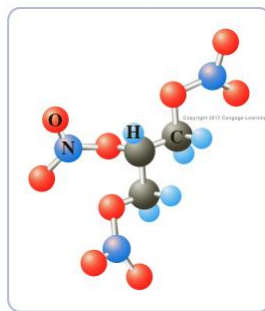
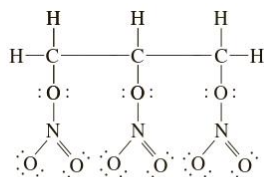


Figure 9.12 ▲

Molecular model of nitroglycerin In this ball-and-stick model, carbon atoms are gray, hydrogen atoms are light blue, oxygen atoms are red, and nitrogen atoms are dark blue.

With just a little jostling, nitroglycerin, $\text{C}_3\text{H}_5(\text{ONO}_2)_3$, can rearrange its atoms to give stable products:



The stability of the products results from their strong bonds, which are much stronger than those in nitroglycerin. Nitrogen, for example, has a strong nitrogen–nitrogen triple bond, and carbon dioxide has two strong carbon–oxygen double bonds. The explosive force of the reaction results from both the rapid reaction and from the large volume increase on forming gaseous products.

In 1867, the Swedish chemist Alfred Nobel discovered that nitroglycerin behaved better when absorbed on diatomaceous earth, a crumbly rock, giving an explosive mixture that Nobel called *dynamite*. Part of the wealth he made from his explosive factories was left in trust to establish the Nobel Prizes. Although explosives are often associated in people's minds with war, they do have many peacetime uses, including road building, mining, and demolition (Figure 9.13).



David Hoffman Photo Library/Alamy

Figure 9.13 ▲

Demolition of a building Explosives are placed at predetermined positions so that when they are detonated, the building collapses on itself.

Example 9.6 Writing Lewis Formulas (Single Bonds Only)

Gaining Mastery Toolbox

Critical Concept 9.6

To write the Lewis electron-dot formula for a molecule, first obtain the total number of valence electrons for the molecule; second, write the skeleton structure, allowing two electrons for each bond between atoms; third, distribute electrons to the outer atoms in order to satisfy the octet rule for each atom; and finally distribute the remaining electrons to the central atom.

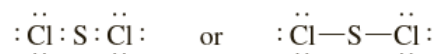
Solution Essentials:

- Skeleton structure of a molecule
- Octet rule
- Number of valence electrons in a main-group atom

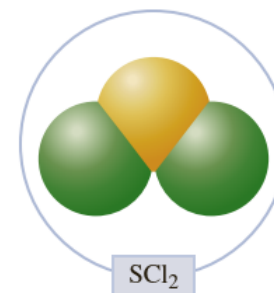
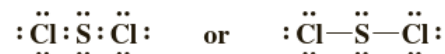
Sulfur dichloride, SCl_2 , is a red, fuming liquid used in the manufacture of insecticides. Write the Lewis formula for the molecule.

Problem Strategy You follow the four steps: (1) Calculate the total number of valence electrons. (2) Write the skeleton structure with two electrons to each bond between atoms. (3) Distribute electrons to the outer atoms to satisfy the octet rule. (4) Distribute the remaining electrons to the central atom.

Solution The number of valence electrons from an atom equals the group number: 6 for S, 7 for each Cl, for a total of 20 electrons. You expect the skeleton structure to have S as the central atom, with the more electronegative Cl atoms bonded to it. After connecting atoms by electron pairs and distributing electrons to the outer atoms, you have



This accounts for 8 electron pairs, or 16 electrons. Subtracting this from the total number of electrons (20) gives 4 electrons, or 2 electron pairs. You place these on the central atom (S). The final Lewis formula is



Answer Check In general, atoms surrounding the central atom have four electron pairs (an octet) about them. Note that our answer follows this “octet” rule. Even the central atom often follows this octet rule, as it does here. (We will see some exceptions to this octet rule for the central atom, however.)

Exercise 9.9 Dichlorodifluoromethane, CCl_2F_2 , is a gas used as a refrigerant and aerosol propellant. Write the Lewis formula for CCl_2F_2 .

See Problems 9.61 and 9.62.

Example 9.7 Writing Lewis Formulas (Including Multiple Bonds)

Gaining Mastery Toolbox

Critical Concept 9.7

If after following the rules for writing an electron-dot formula, you find that the central atom does not have enough electrons for an octet, transfer electron pairs from outer atoms to form multiple bonds in order to give an octet to the central atom.

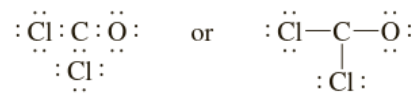
Solution Essentials:

- Skeleton structure of a molecule
- Lewis representation of multiple bonds
- Octet rule
- Number of valence electrons in a main-group atom

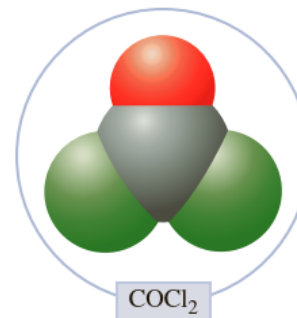
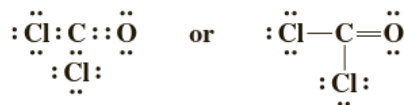
Carbonyl chloride, or phosgene, COCl_2 , is a highly toxic gas used as a starting material for the preparation of polyurethane plastics. What is the electron-dot formula of COCl_2 ?

Problem Strategy You follow the four steps as in the previous example, but after distributing the remaining electrons to the central atom, you find that it does not have enough electrons to give an octet. To rectify this, you transfer an electron pair from an outer atom to form a double bond (part of Step 4). Do you move the electron pair from the Cl or the O atom? It helps to remember that C, N, O, and S atoms often form double bonds. In this case, you move the electron pair from the O atom.

Solution The total number of valence electrons is $4 + 6 + (2 \times 7) = 24$. You expect the more electropositive atom, C, to be central, with the O and Cl atoms bonded to it. After distributing electron pairs to these surrounding atoms, you have



This accounts for all 24 valence electrons, leaving only 6 electrons on C. This is two fewer than an octet, so you move a pair of electrons on the O atom to give a carbon–oxygen double bond. The electron-dot formula of COCl_2 is



Answer Check Note that the atoms surrounding the central atom have four electron pairs about them; that is, they follow the octet rule. The central atom also follows the octet rule, although exceptions occur.

Exercise 9.10 Write the electron-dot formula of carbon dioxide, CO_2 .

See Problems 9.63 and 9.64.

Example 9.8 Writing Lewis Formulas (Ionic Species)

Gaining Mastery Toolbox

Critical Concept 9.8

In obtaining the Lewis formula for an ion, note that the total number of valence electrons equals that for the neutral molecule less the charge on the ion (that is, subtract if a positive ion and add if a negative ion).

Solution Essentials:

- Skeleton structure of a molecule
- Lewis representation of multiple bonds
- Octet rule
- Number of valence electrons in a main-group atom

Obtain the electron-dot formula (Lewis formula) of the BF_4^- ion.

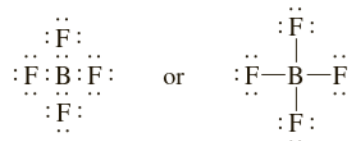
Problem Strategy After calculating the total number of valence electrons for the neutral atoms, you add or subtract electrons to account for the ion charge. If the ion is negative, add electrons to give the negative charge. If the ion is positive, subtract electrons to give the positive charge. Once you have the total number of valence electrons in the ion, proceed with the rest of the steps to write the electron-dot formula.

Solution The total valence electrons provided by the boron and four fluorine atoms is $3 + (4 \times 7) = 31$. Because the anion has a charge of -1 , it has one more electron than is provided by the neutral atoms. Thus, the total number of valence electrons is 32 (or 16 electron pairs). You assume that the skeleton structure has boron as the central

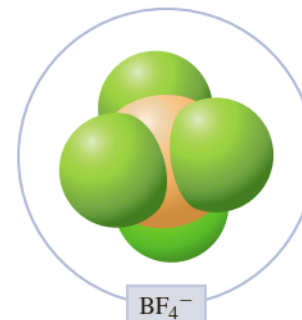
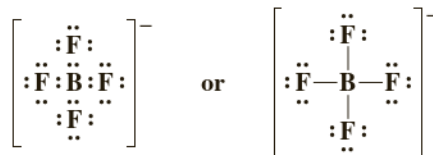
(continued)

Example 9.8 (continued)

atom, with the more electronegative F atoms bonded to it. After connecting the B and F atoms by bonds and placing electron pairs around the F atoms to satisfy the octet rule, you obtain



This uses up all 32 electrons. The charge on the entire ion is indicated by the minus sign written as a superscript to square brackets enclosing the electron-dot formula.



10.3 Valence Bond Theory

The VSEPR model, based largely on Lewis structures, provides a relatively simple and straightforward method for predicting the geometry of molecules. But as we noted earlier, the Lewis theory of chemical bonding does not clearly explain why chemical bonds exist. Relating the formation of a covalent bond to the pairing of electrons was a step in the right direction, but it did not go far enough. For example, the Lewis theory describes the single bond between the H atoms in H_2 and that between the F atoms in F_2 in essentially the same way—as the pairing of two electrons. Yet these two molecules have quite different bond enthalpies and bond lengths (436.4 kJ/mol and 74 pm for H_2 and

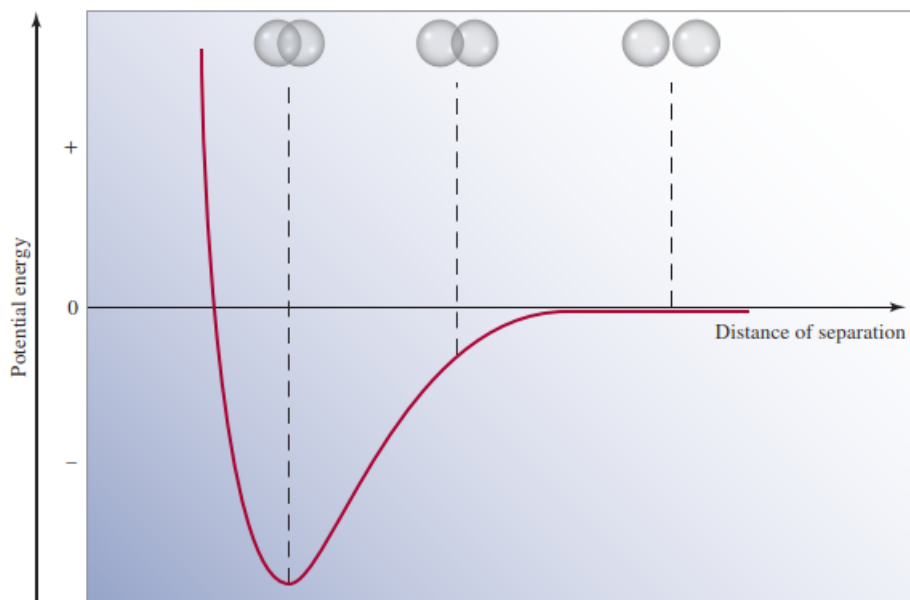


Figure 10.5 Change in potential energy of two H atoms with their distance of separation. At the point of minimum potential energy, the H_2 molecule is in its most stable state and the bond length is 74 pm. The spheres represent the 1s orbitals.

150.6 kJ/mol and 142 pm for F_2). These and many other facts cannot be explained by the Lewis theory. For a more complete explanation of chemical bond formation we look to quantum mechanics. In fact, the quantum mechanical study of chemical bonding also provides a means for understanding molecular geometry.

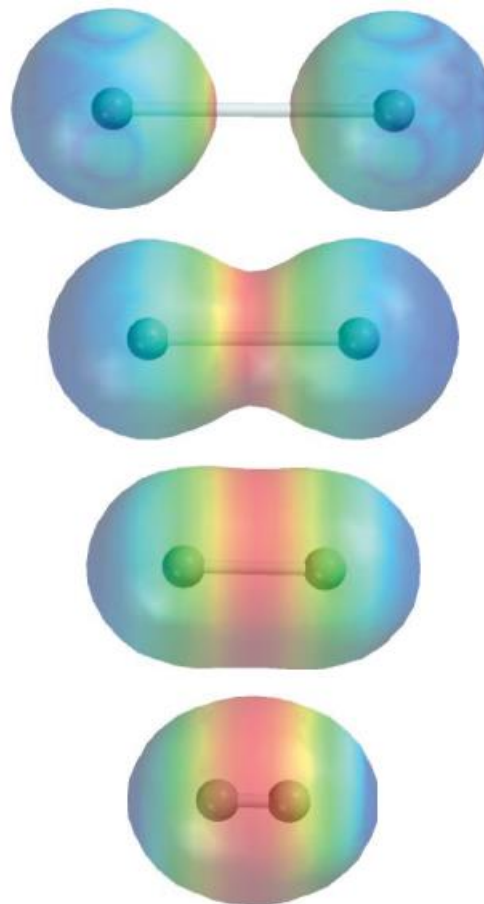
At present, two quantum mechanical theories are used to describe covalent bond formation and the electronic structure of molecules. *Valence bond (VB) theory* assumes that the electrons in a molecule occupy atomic orbitals of the individual atoms. It enables us to retain a picture of individual atoms taking part in the bond formation. The second theory, called *molecular orbital (MO) theory*, assumes the formation of molecular orbitals from the atomic orbitals. Neither theory perfectly explains all aspects of bonding, but each has contributed something to our understanding of many observed molecular properties.

Let us start our discussion of valence bond theory by considering the formation of a H_2 molecule from two H atoms. The Lewis theory describes the H—H bond in terms of the pairing of the two electrons on the H atoms. In the framework of valence bond theory, the covalent H—H bond is formed by the *overlap* of the two $1s$ orbitals in the H atoms. By overlap, we mean that the two orbitals share a common region in space.

What happens to two H atoms as they move toward each other and form a bond? Initially, when the two atoms are far apart, there is no interaction. We say that the potential energy of this system (that is, the two H atoms) is zero. As the atoms approach each other, each electron is attracted by the nucleus of the other atom; at the same time, the electrons repel each other, as do the nuclei. While the atoms are still separated, attraction is stronger than repulsion, so that the potential energy of the system *decreases* (that is, it becomes negative) as the atoms approach each other (Figure 10.5). This trend continues until the potential energy reaches a minimum value. At this point, when the system has the lowest potential energy, it is most stable. This condition corresponds to substantial overlap of the $1s$ orbitals and the formation of a stable H_2 molecule. If the distance between nuclei were to decrease further, the potential energy would rise steeply and finally become positive as a result of the increased electron-electron and nuclear-nuclear repulsions. In accord with the law of conservation of energy, the decrease in potential energy as a result of H_2 formation must be accompanied by a release of energy. Experiments show that as a H_2 molecule is formed from two H atoms, heat is given off. The converse is also true. To break a H—H bond, energy must be supplied to the molecule. Figure 10.6 is another way of viewing the formation of a H_2 molecule.

Recall that an object has potential energy by virtue of its position.

Figure 10.6 Top to bottom: As two H atoms approach each other, their 1s orbitals begin to interact and each electron begins to feel the attraction of the other proton. Gradually, the electron density builds up in the region between the two nuclei (red color). Eventually, a stable H_2 molecule is formed with an internuclear distance of 74 pm.



Thus, valence bond theory gives a clearer picture of chemical bond formation than the Lewis theory does. Valence bond theory states that a stable molecule forms from reacting atoms when the potential energy of the system has decreased to a minimum; the Lewis theory ignores energy changes in chemical bond formation.

The concept of overlapping atomic orbitals applies equally well to diatomic molecules other than H_2 . Thus, a stable F_2 molecule forms when the $2p$ orbitals (containing the unpaired electrons) in the two F atoms overlap to form a covalent bond. Similarly, the formation of the HF molecule can be explained by the overlap of the $1s$ orbital in H with the $2p$ orbital in F. In each case, VB theory accounts for the changes in potential energy as the distance between the reacting atoms changes. Because the orbitals involved are not the same kind in all cases, we can see why the bond enthalpies and bond lengths in H_2 , F_2 , and HF might be different. As we stated earlier, Lewis theory treats *all* covalent bonds the same way and offers no explanation for the differences among covalent bonds.

The orbital diagram of the F atom is shown in Section 7.8.

Limitations of Valence Bond Theory:

1.No Explanation of Delocalization:

1. Fails to explain delocalized bonding in molecules like benzene (C_6H_6), where π -electrons are shared over multiple atoms.

2.Inability to Explain Magnetic Behavior:

1. Cannot accurately predict the paramagnetism of O_2 , which has unpaired electrons (explained better by Molecular Orbital Theory).

3.Does Not Explain Bond Energies or Lengths Accurately:

1. Quantitative predictions of bond strength and bond length are often inaccurate or inconsistent.

4.Cannot Explain Color or Spectra:

1. Fails to account for absorption/emission spectra, especially in transition metal complexes.

5.No Consideration of Molecular Orbitals:

1. Ignores the concept of molecular orbitals, which are essential in understanding bonding in molecules with more complex electron interactions.

10.4 Hybridization of Atomic Orbitals

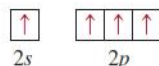
The concept of atomic orbital overlap should apply also to polyatomic molecules. However, a satisfactory bonding scheme must account for molecular geometry. We will discuss three examples of VB treatment of bonding in polyatomic molecules.

sp^3 Hybridization

Consider the CH_4 molecule. Focusing only on the valence electrons, we can represent the orbital diagram of C as

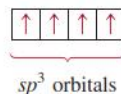


Because the carbon atom has two unpaired electrons (one in each of the two $2p$ orbitals), it can form only two bonds with hydrogen in its ground state. Although the species CH_2 is known, it is very unstable. To account for the four C—H bonds in methane, we can try to promote (that is, energetically excite) an electron from the $2s$ orbital to the $2p$ orbital:



Now there are four unpaired electrons on C that could form four C—H bonds. However, the geometry is wrong, because three of the HCH bond angles would have to be 90° (remember that the three $2p$ orbitals on carbon are mutually perpendicular), and yet *all* HCH angles are 109.5° .

To explain the bonding in methane, VB theory uses hypothetical **hybrid orbitals**, which are *atomic orbitals obtained when two or more nonequivalent orbitals of the same atom combine in preparation for covalent bond formation*. **Hybridization** is the term applied to *the mixing of atomic orbitals in an atom (usually a central atom) to generate a set of hybrid orbitals*. We can generate four equivalent hybrid orbitals for carbon by mixing the $2s$ orbital and the three $2p$ orbitals:



Because the new orbitals are formed from one s and three p orbitals, they are called sp^3 hybrid orbitals. Figure 10.7 shows the shape and orientations of the sp^3 orbitals. These four hybrid orbitals are directed toward the four corners of a regular tetrahedron.

Video
Hybridization

sp^3 is pronounced "s-p three."

Video
Molecular Shape and Orbital Hybridization

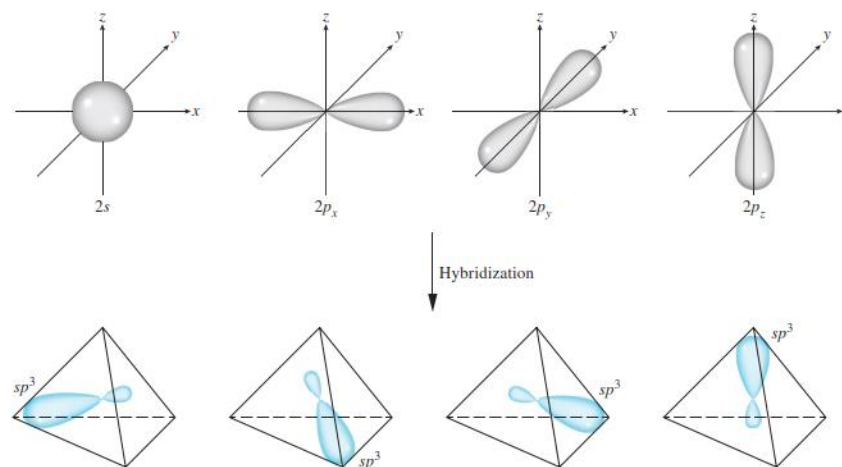


Figure 10.7 Formation of four sp^3 hybrid orbitals from one $2s$ and three $2p$ orbitals. The sp^3 orbitals point to the corners of a tetrahedron.

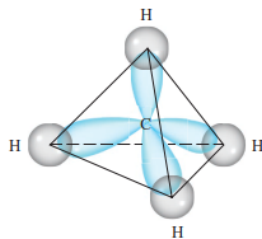


Figure 10.8 Formation of four bonds between the carbon sp^3 hybrid orbitals and the hydrogen $1s$ orbitals in CH_4 . The smaller lobes are not shown.

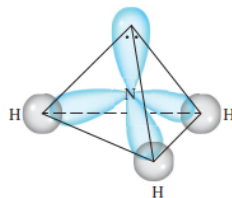
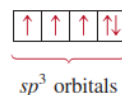


Figure 10.9 The sp^3 -hybridized N atom in NH_3 . Three sp^3 hybrid orbitals form bonds with the H atoms. The fourth is occupied by nitrogen's lone pair.

Figure 10.8 shows the formation of four covalent bonds between the carbon sp^3 hybrid orbitals and the hydrogen $1s$ orbitals in CH_4 . Thus CH_4 has a tetrahedral shape, and all the HCH angles are 109.5° . Note that although energy is required to bring about hybridization, this input is more than compensated for by the energy released upon the formation of C—H bonds. (Recall that bond formation is an exothermic process.)

The following analogy is useful for understanding hybridization. Suppose that we have a beaker of a red solution and three beakers of blue solutions and that the volume of each is 50 mL. The red solution corresponds to one $2s$ orbital, the blue solutions represent three $2p$ orbitals, and the four equal volumes symbolize four separate orbitals. By mixing the solutions we obtain 200 mL of a purple solution, which can be divided into four 50-mL portions (that is, the hybridization process generates four sp^3 orbitals). Just as the purple color is made up of the red and blue components of the original solutions, the sp^3 hybrid orbitals possess both s and p orbital characteristics.

Another example of sp^3 hybridization is ammonia (NH_3). Table 10.1 shows that the arrangement of four electron pairs is tetrahedral, so that the bonding in NH_3 can be explained by assuming that N, like C in CH_4 , is sp^3 -hybridized. The ground-state electron configuration of N is $1s^2 2s^2 2p^3$, so that the orbital diagram for the sp^3 -hybridized N atom is



Three of the four hybrid orbitals form covalent N—H bonds, and the fourth hybrid orbital accommodates the lone pair on nitrogen (Figure 10.9). Repulsion between the lone-pair electrons and electrons in the bonding orbitals decreases the HNH bond angles from 109.5° to 107.3° .

It is important to understand the relationship between hybridization and the VSEPR model. We use hybridization to describe the bonding scheme only when the arrangement of electron pairs has been predicted using VSEPR. If the VSEPR model predicts a tetrahedral arrangement of electron pairs, then we assume that one s and three p orbitals are hybridized to form four sp^3 hybrid orbitals. The following are examples of other types of hybridization.

sp Hybridization

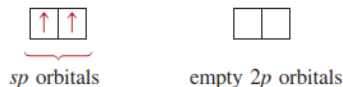
The beryllium chloride ($BeCl_2$) molecule is predicted to be linear by VSEPR. The orbital diagram for the valence electrons in Be is



We know that in its ground state, Be does not form covalent bonds with Cl because its electrons are paired in the $2s$ orbital. So we turn to hybridization for an explanation of Be's bonding behavior. First, we promote a $2s$ electron to a $2p$ orbital, resulting in



Now there are two Be orbitals available for bonding, the $2s$ and $2p$. However, if two Cl atoms were to combine with Be in this excited state, one Cl atom would share a $2s$ electron and the other Cl would share a $2p$ electron, making two nonequivalent BeCl bonds. This scheme contradicts experimental evidence. In the actual $BeCl_2$ molecule, the two BeCl bonds are identical in every respect. Thus, the $2s$ and $2p$ orbitals must be mixed, or hybridized, to form two equivalent sp hybrid orbitals:



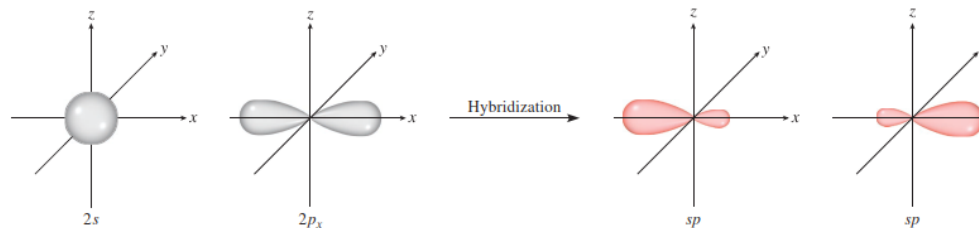


Figure 10.10 Formation of sp hybrid orbitals.

Figure 10.10 shows the shape and orientation of the sp orbitals. These two hybrid orbitals lie on the same line, the x -axis, so that the angle between them is 180° . Each of the BeCl bonds is then formed by the overlap of a $\text{Be } sp$ hybrid orbital and a $\text{Cl } 3p$ orbital, and the resulting BeCl_2 molecule has a linear geometry (Figure 10.11).



Figure 10.11 The linear geometry of BeCl_2 can be explained by assuming that Be sp -hybridized. The two sp hybrid orbitals overlap with the two chlorine $3p$ orbitals to form two covalent bonds.

sp^2 Hybridization

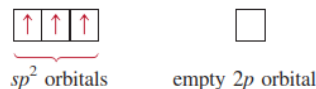
Next we will look at the BF_3 (boron trifluoride) molecule, known to have a planar geometry. Considering only the valence electrons, the orbital diagram of B is



First, we promote a $2s$ electron to an empty $2p$ orbital:



Mixing the $2s$ orbital with the two $2p$ orbitals generates three sp^2 hybrid orbitals:



sp^2 is pronounced "s-p two."

These three sp^2 orbitals lie in the same plane, and the angle between any two of them is 120° (Figure 10.12). Each of the BF bonds is formed by the overlap of a boron sp^2

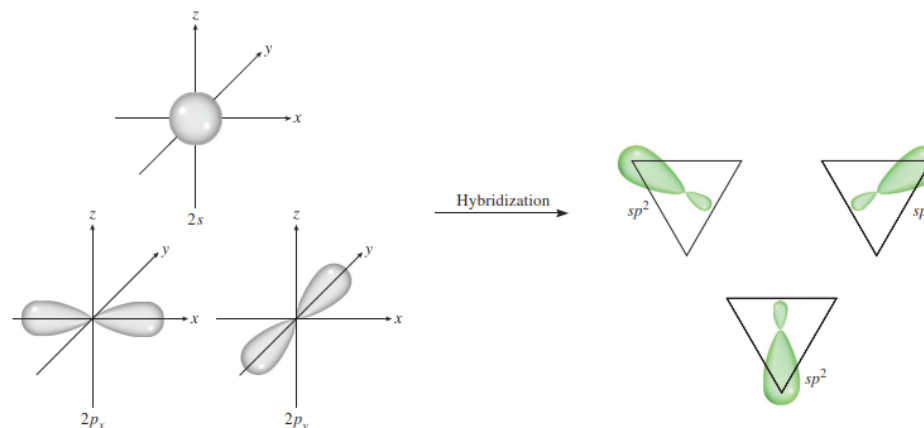


Figure 10.12 Formation of sp^2 hybrid orbitals.

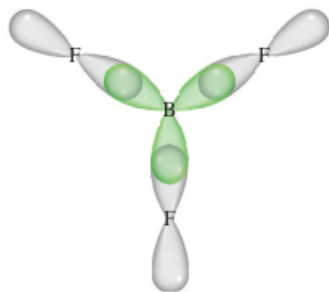


Figure 10.13 The sp^2 hybrid orbitals of boron overlap with the $2p$ orbitals of fluorine. The BF_3 molecule is planar, and all the FBF angles are 120° .

hybrid orbital and a fluorine $2p$ orbital (Figure 10.13). The BF_3 molecule is planar with all the FBF angles equal to 120° . This result conforms to experimental findings and also to VSEPR predictions.

You may have noticed an interesting connection between hybridization and the octet rule. Regardless of the type of hybridization, an atom starting with one s and three p orbitals would still possess four orbitals, enough to accommodate a total of eight electrons in a compound. For elements in the second period of the periodic table, eight is the maximum number of electrons that an atom of any of these elements can accommodate in the valence shell. This is the reason that the octet rule is usually obeyed by the second-period elements.

The situation is different for an atom of a third-period element. If we use only the $3s$ and $3p$ orbitals of the atom to form hybrid orbitals in a molecule, then the octet rule applies. However, in some molecules the same atom may use one or more $3d$ orbitals, in addition to the $3s$ and $3p$ orbitals, to form hybrid orbitals. In these cases, the octet rule does not hold. We will see specific examples of the participation of the $3d$ orbital in hybridization shortly.

To summarize our discussion of hybridization, we note that

- The concept of hybridization is not applied to isolated atoms. It is a theoretical model used only to explain covalent bonding.
- Hybridization is the mixing of at least two nonequivalent atomic orbitals, for example, s and p orbitals. Therefore, a hybrid orbital is not a pure atomic orbital. Hybrid orbitals and pure atomic orbitals have very different shapes.
- The number of hybrid orbitals generated is equal to the number of pure atomic orbitals that participate in the hybridization process.
- Hybridization requires an input of energy; however, the system more than recovers this energy during bond formation.
- Covalent bonds in polyatomic molecules and ions are formed by the overlap of hybrid orbitals, or of hybrid orbitals with unhybridized ones. Therefore, the hybridization bonding scheme is still within the framework of valence bond theory; electrons in a molecule are assumed to occupy hybrid orbitals of the individual atoms.

Table 10.4 summarizes sp , sp^2 , and sp^3 hybridization (as well as other types that we will discuss shortly).

Procedure for Hybridizing Atomic Orbitals

Before going on to discuss the hybridization of d orbitals, let us specify what we need to know in order to apply hybridization to bonding in polyatomic molecules in general. In essence, hybridization simply extends Lewis theory and the VSEPR model. To assign a suitable state of hybridization to the central atom in a molecule, we must have some idea about the geometry of the molecule. The steps are as follows:

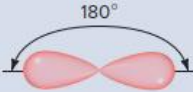
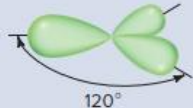
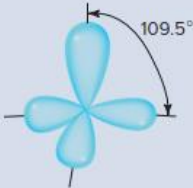
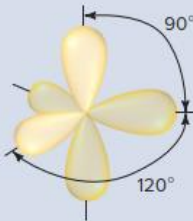
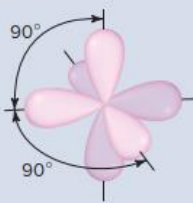
1. Draw the Lewis structure of the molecule.
2. Predict the overall arrangement of the electron pairs (both bonding pairs and lone pairs) using the VSEPR model (see Table 10.1).
3. Deduce the hybridization of the central atom by matching the arrangement of the electron pairs with those of the hybrid orbitals shown in Table 10.4.



Student Hot Spot

Student data indicate you may struggle with hybridization. Access your eBook for additional Learning Resources on this topic.

Table 10.4 Important Hybrid Orbitals and Their Shapes

Pure Atomic Orbitals of the Central Atom	Hybridization of the Central Atom	Number of Hybrid Orbitals	Shape of Hybrid Orbitals	Examples
s, p	sp	2	 Linear	BeCl_2
s, p, p	sp^2	3	 Trigonal planar	BF_3
s, p, p, p	sp^3	4	 Tetrahedral	$\text{CH}_4, \text{NH}_4^+$
s, p, p, p, d	sp^3d	5	 Trigonal bipyramidal	PCl_5
s, p, p, p, d, d	sp^3d^2	6	 Octahedral	SF_6

Example 10.3

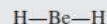
Determine the hybridization state of the central (underlined) atom in each of the following molecules: (a) $\underline{\text{Be}}\text{H}_2$, (b) $\underline{\text{Al}}\text{I}_3$, (c) $\underline{\text{P}}\text{F}_3$. Describe the hybridization process and determine the molecular geometry in each case.

Strategy The steps for determining the hybridization of the central atom in a molecule are

draw Lewis structure of the molecule	→	use VSEPR to determine the electron pair arrangement surrounding the central atom (Table 10.1)	→	use Table 10.4 to determine the hybridization state of the central atom
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Solution

- (a) The ground-state electron configuration of Be is $1s^2 2s^2$ and the Be atom has two valence electrons. The Lewis structure of BeH_2 is



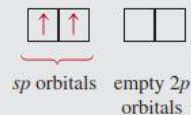
There are two bonding pairs around Be; therefore, the electron pair arrangement is linear. We conclude that Be uses sp hybrid orbitals in bonding with H, because sp orbitals have a linear arrangement (see Table 10.4). The hybridization process can be imagined as follows. First, we draw the orbital diagram for the ground state of Be:



By promoting a $2s$ electron to the $2p$ orbital, we get the excited state:

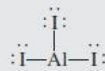


The $2s$ and $2p$ orbitals then mix to form two hybrid orbitals:

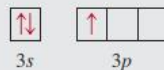


The two $\text{Be}—\text{H}$ bonds are formed by the overlap of the Be sp orbitals with the $1s$ orbitals of the H atoms. Thus, BeH_2 is a linear molecule.

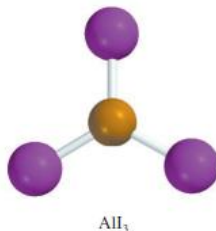
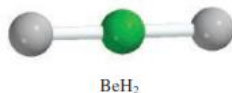
- (b) The ground-state electron configuration of Al is $[\text{Ne}]3s^2 3p^1$. Therefore, the Al atom has three valence electrons. The Lewis structure of AlI_3 is



There are three pairs of electrons around Al; therefore, the electron pair arrangement is trigonal planar. We conclude that Al uses sp^2 hybrid orbitals in bonding with I because sp^2 orbitals have a trigonal planar arrangement (see Table 10.4). The orbital diagram of the ground-state Al atom is



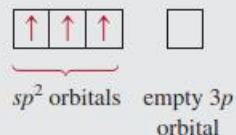
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By promoting a 3s electron into the 3p orbital we obtain the following excited state:

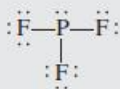


The 3s and two 3p orbitals then mix to form three sp^2 hybrid orbitals:



The sp^2 hybrid orbitals overlap with the 5p orbitals of I to form three covalent Al—I bonds. We predict that the AlI_3 molecule is trigonal planar and all the IAlI angles are 120° .

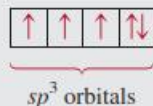
- (c) The ground-state electron configuration of P is $[Ne]3s^23p^3$. Therefore, P atom has five valence electrons. The Lewis structure of PF_3 is



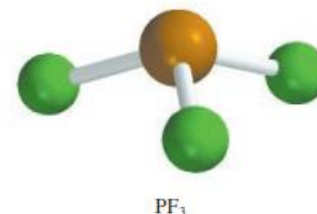
There are four pairs of electrons around P; therefore, the electron pair arrangement is tetrahedral. We conclude that P uses sp^3 hybrid orbitals in bonding to F, because sp^3 orbitals have a tetrahedral arrangement (see Table 10.4). The hybridization process can be imagined to take place as follows. The orbital diagram of the ground-state P atom is



By mixing the 3s and 3p orbitals, we obtain four sp^3 hybrid orbitals.



As in the case of NH_3 , one of the sp^3 hybrid orbitals is used to accommodate the lone pair on P. The other three sp^3 hybrid orbitals form covalent P—F bonds with the 2p orbitals of F. We predict the geometry of the molecule to be trigonal pyramidal; the FPF angle should be somewhat less than 109.5° .



PF_3

Practice Exercise Determine the hybridization state of the underlined atoms in the following compounds: (a) $\underline{S}iBr_4$, (b) $\underline{B}Cl_3$.

Similar problems: 10.31, 10.33.

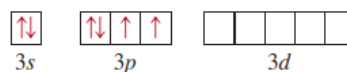
Hybridization of s , p , and d Orbitals

We have seen that hybridization neatly explains bonding that involves s and p orbitals. For elements in the third period and beyond, however, we cannot always account for molecular geometry by assuming that only s and p orbitals hybridize. To understand the formation of molecules with trigonal bipyramidal and octahedral geometries, for instance, we must include d orbitals in the hybridization concept.

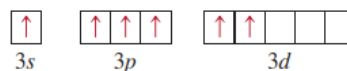
Consider the SF_6 molecule as an example. In Section 10.1 we saw that this molecule has octahedral geometry, which is also the arrangement of the six electron pairs. Table 10.4 shows that the S atom is sp^3d^2 -hybridized in SF_6 . The ground-state electron



configuration of S is $[\text{Ne}]3s^23p^4$. Focusing only on the valence electrons, we have the orbital diagram

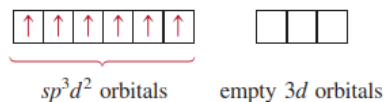


Because the $3d$ level is quite close in energy to the $3s$ and $3p$ levels, we can promote $3s$ and $3p$ electrons to two of the $3d$ orbitals:



sp^3d^2 is pronounced "s-p three d two."

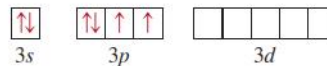
Mixing the $3s$, three $3p$, and two $3d$ orbitals generates six sp^3d^2 hybrid orbitals:



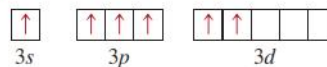
The six S—F bonds are formed by the overlap of the hybrid orbitals of the S atom with the $2p$ orbitals of the F atoms. Because there are 12 electrons around the S atom, the octet rule is violated. The use of d orbitals in addition to s and p orbitals to form an expanded octet (see Section 9.9) is an example of *valence-shell expansion*. Second-period elements, unlike third-period elements, do not have $2d$ energy levels, so they can never expand their valence shells. (Recall that when $n = 2$, $l = 0$ and 1. Thus, we can only have $2s$ and $2p$ orbitals.) Hence atoms of second-period elements can never be surrounded by more than eight electrons in any of their compounds.

Example 10.4 deals with valence-shell expansion in a third-period element.

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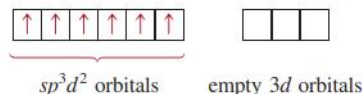


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sp^3d^2 is pronounced "s-p three d two."

Mixing the $3s$, three $3p$, and two $3d$ orbitals generates six sp^3d^2 hybrid orbitals:



The six S—F bonds are formed by the overlap of the hybrid orbitals of the S atom with the $2p$ orbitals of the F atoms. Because there are 12 electrons around the S atom, the octet rule is violated. The use of d orbitals in addition to s and p orbitals to form an expanded octet (see Section 9.9) is an example of *valence-shell expansion*. Second-period elements, unlike third-period elements, do not have $2d$ energy levels, so they can never expand their valence shells. (Recall that when $n = 2$, $l = 0$ and 1. Thus, we can only have $2s$ and $2p$ orbitals.) Hence atoms of second-period elements can never be surrounded by more than eight electrons in any of their compounds.

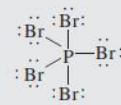
Example 10.4 deals with valence-shell expansion in a third-period element.

Example 10.4

Describe the hybridization state of phosphorus in phosphorus pentabromide (PBr_5).

Strategy Follow the same procedure shown in Example 10.3.

Solution The ground-state electron configuration of P is $[\text{Ne}]3s^23p^3$. Therefore, the P atom has five valence electrons. The Lewis structure of PBr_5 is



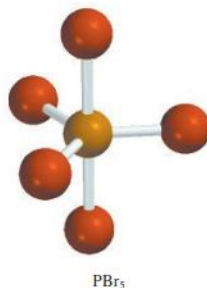
There are five pairs of electrons around P; therefore, the electron pair arrangement is trigonal bipyramidal. We conclude that P uses sp^3d hybrid orbitals in bonding to Br, because sp^3d hybrid orbitals have a trigonal bipyramidal arrangement (see Table 10.4). The hybridization process can be imagined as follows. The orbital diagram of the ground-state P atom is



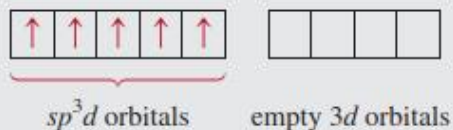
Promoting a $3s$ electron into a $3d$ orbital results in the following excited state:



(Continued)



Mixing the one $3s$, three $3p$, and one $3d$ orbitals generates five sp^3d hybrid orbitals:



These hybrid orbitals overlap the $4p$ orbitals of Br to form five covalent P—Br bonds. Because there are no lone pairs on the P atom, the geometry of PBr_5 is trigonal bipyramidal.

Practice Exercise Describe the hybridization state of Se in SeF_6 .

Similar problem: 10.36.

10.1 Molecular Geometry

Molecular geometry is the three-dimensional arrangement of atoms in a molecule. A molecule's geometry affects its physical and chemical properties, such as melting point, boiling point, density, and the types of reactions it undergoes. In general, bond lengths and bond angles must be determined by experiment. However, there is a simple procedure that enables us to predict with considerable success the overall geometry of a molecule or ion if we know the number of electrons surrounding a central atom in its Lewis structure. The basis of this approach is the assumption that electron pairs in the valence shell of an atom repel one another. The **valence shell** is *the outermost electron-occupied shell of an atom; it holds the electrons that are usually involved in bonding*. In a covalent bond, a pair of electrons, often called the *bonding pair*, is responsible for holding two atoms together. However, in a polyatomic molecule, where there are two or more bonds between the central atom and the surrounding atoms, the repulsion between electrons in different bonding pairs causes them to remain as far apart as possible. The geometry that the molecule ultimately assumes (as defined by the positions of all the atoms) minimizes the repulsion. This approach to the study of molecular geometry is called the **valence-shell electron-pair repulsion (VSEPR) model**, because *it accounts*

The term "central atom" means an atom that is not a terminal atom in a polyatomic molecule.

VSEPR is pronounced "vesper."

for the geometric arrangements of electron pairs around a central atom in terms of the electrostatic repulsion between electron pairs.

Two general rules govern the use of the VSEPR model:

1. As far as electron-pair repulsion is concerned, double bonds and triple bonds can be treated like single bonds. This approximation is good for qualitative purposes. However, you should realize that in reality multiple bonds are “larger” than single bonds; that is, because there are two or three bonds between two atoms, the electron density occupies more space.
2. If a molecule has two or more resonance structures, we can apply the VSEPR model to any one of them. Formal charges are usually not shown.

With this model in mind, we can predict the geometry of molecules (and ions) in a systematic way. For this purpose, it is convenient to divide molecules into two categories, according to whether or not the central atom has lone pairs.

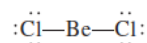
Molecules in Which the Central Atom Has No Lone Pairs

For simplicity we will consider molecules that contain atoms of only two elements, A and B, of which A is the central atom. These molecules have the general formula AB_x , where x is an integer 2, 3, . . . (If $x = 1$, we have the diatomic molecule AB, which is linear by definition.) In the vast majority of cases, x is between 2 and 6.

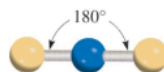
Table 10.1 shows five possible arrangements of electron pairs around the central atom A. As a result of mutual repulsion, the electron pairs stay as far from one another as possible. Note that the table shows arrangements of the electron pairs but not the positions of the atoms that surround the central atom. Molecules in which the central atom has no lone pairs have one of these five arrangements of bonding pairs. Using Table 10.1 as a reference, let us take a close look at the geometry of molecules with the formulas AB_2 , AB_3 , AB_4 , AB_5 , and AB_6 .

AB_2 : Beryllium Chloride ($BeCl_2$)

The Lewis structure of beryllium chloride in the gaseous state is

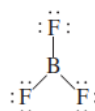


Because the bonding pairs repel each other, they must be at opposite ends of a straight line in order for them to be as far apart as possible. Thus, the $ClBeCl$ angle is predicted to be 180° , and the molecule is linear (see Table 10.1). The “ball-and-stick” model of $BeCl_2$ is



AB_3 : Boron Trifluoride (BF_3)

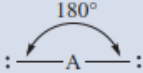
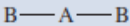
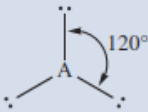
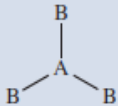
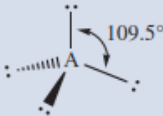
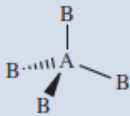
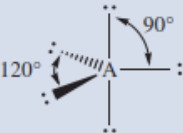
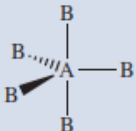
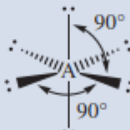
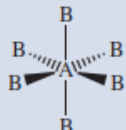
Boron trifluoride contains three covalent bonds, or bonding pairs.



In the most stable arrangement, the three BF bonds point to the corners of an equilateral triangle with B in the center of the triangle. According to Table 10.1, the geometry of

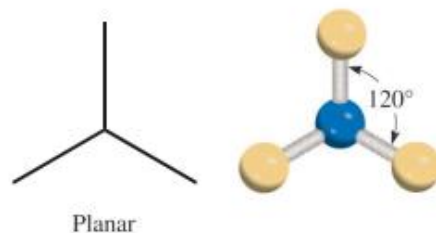
Table 10.1

Arrangement of Electron Pairs About a Central Atom (A) in a Molecule and Geometry of Some Simple Molecules and Ions in Which the Central Atom Has No Lone Pairs

Number of Electron Pairs	Arrangement of Electron Pairs*	Molecular Geometry*	Examples
2	 Linear	 Linear	BeCl ₂ , HgCl ₂
3	 Trigonal planar	 Trigonal planar	BF ₃
4	 Tetrahedral	 Tetrahedral	CH ₄ , NH ₄ ⁺
5	 Trigonal bipyramidal	 Trigonal bipyramidal	PCl ₅
6	 Octahedral	 Octahedral	SF ₆

*Bonds coming out of the page are represented as solid wedges. Bonds going into the page are represented as dashed wedges. Bonds in the plane of the page are represented as solid lines.

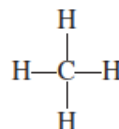
BF_3 is trigonal planar because all four atoms lie in the same plane and the three end atoms form an equilateral triangle:



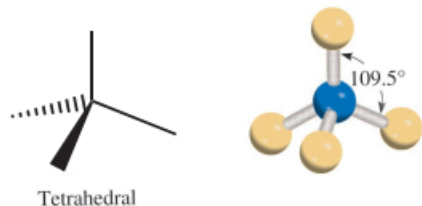
Thus, each of the three FBF angles is 120° .

AB₄: Methane (CH_4)

The Lewis structure of methane is

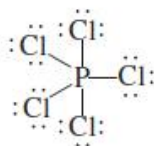


Because there are four bonding pairs, the geometry of CH_4 is tetrahedral (see Table 10.1). A *tetrahedron* has four sides (the prefix *tetra-* means “four”), or faces, all of which are equilateral triangles. In a tetrahedral molecule, the central atom (C in this case) is located at the center of the tetrahedron and the other four atoms are at the corners. The bond angles are all 109.5° .

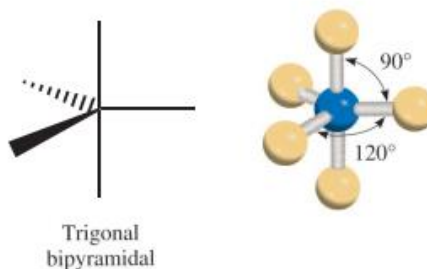


AB₅: Phosphorus Pentachloride (PCl₅)

The Lewis structure of phosphorus pentachloride (in the gas phase) is



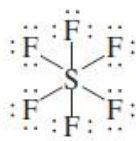
The only way to minimize the repulsive forces among the five bonding pairs is to arrange the PCl bonds in the form of a trigonal bipyramid (see Table 10.1). A trigonal bipyramid can be generated by joining two tetrahedrons along a common triangular base:



The central atom (P in this case) is at the center of the common triangle with the surrounding atoms positioned at the five corners of the trigonal bipyramid. The atoms that are above and below the triangular plane are said to occupy *axial* positions, and those that are in the triangular plane are said to occupy *equatorial* positions. The angle between any two equatorial bonds is 120°; that between an axial bond and an equatorial bond is 90°, and that between the two axial bonds is 180°.

AB₆: Sulfur Hexafluoride (SF₆)

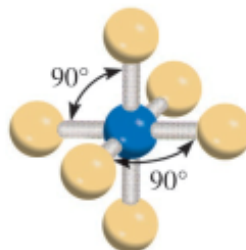
The Lewis structure of sulfur hexafluoride is



The most stable arrangement of the six SF bonding pairs is in the shape of an octahedron, as shown in Table 10.1. An octahedron has eight sides (the prefix *octa-* means “eight”). It can be generated by joining two square pyramids on a common base. The central atom (S in this case) is at the center of the square base and the surrounding atoms are at the six corners. All bond angles are 90° except the one made by the bonds between the central atom and the pairs of atoms that are diametrically opposite each other. That angle is 180° . Because the six bonds are equivalent in an octahedral molecule, we cannot use the terms “axial” and “equatorial” as in a trigonal bipyramidal molecule.



Octahedral



Hydrogen Bonding

It is interesting to compare fluoromethane, CH_3F , and methanol, CH_3OH (Figure 11.23). They have about the same molecular weight (34 for CH_3F and 32 for CH_3OH) and about the same dipole moment (1.81 D for CH_3F and 1.70 D for CH_3OH). You might expect these substances to have about the same intermolecular attractive forces and therefore about the same boiling points. In fact, the boiling points are quite different. Fluoromethane boils at -78°C and is a gas under normal conditions. Methanol boils at 65°C and is normally a liquid. We must conclude that there is an intermolecular attraction in methanol that is not present in fluoromethane.

We have already seen that the properties of water and glycerol cannot be explained in terms of van der Waals forces alone. What water, glycerol, and methanol have in common is one or more —OH groups.

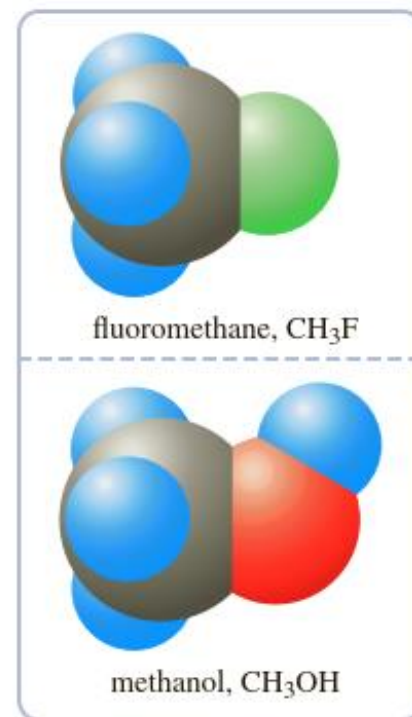
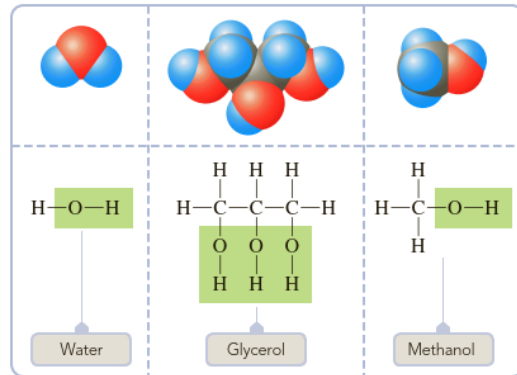


Figure 11.23 ▲

Fluoromethane and methanol Space-filling molecular models.

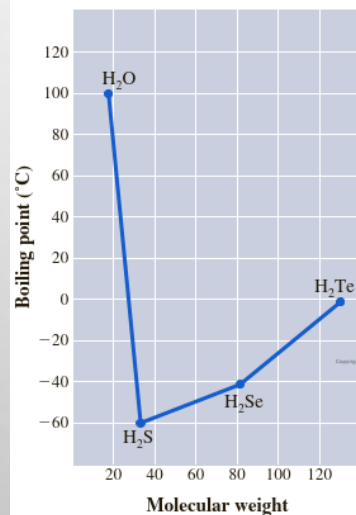


Molecules that have the —OH group are subject to an additional attractive force called hydrogen bonding. **Hydrogen bonding** is a weak to moderate attractive force that exists between a hydrogen atom covalently bonded to a very electronegative atom, *X*, and a lone pair of electrons on another small, electronegative atom, *Y*. Hydrogen bonding is represented in structural drawings by a series of dots.

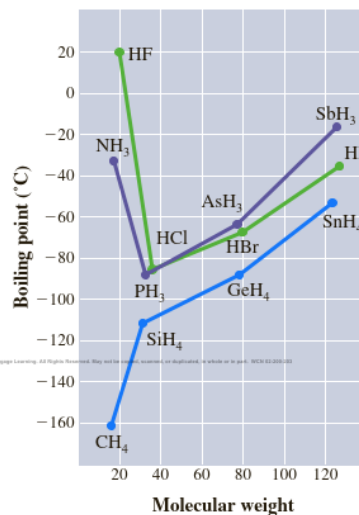


Usually, hydrogen bonding is seen in cases where X and Y are the atoms F, O, or N.

Figure 11.24a shows a plot of boiling point versus molecular weight for the hydrides (binary compounds with hydrogen) of the Group 6A elements. If London forces were the only intermolecular forces present in this series of compounds, the



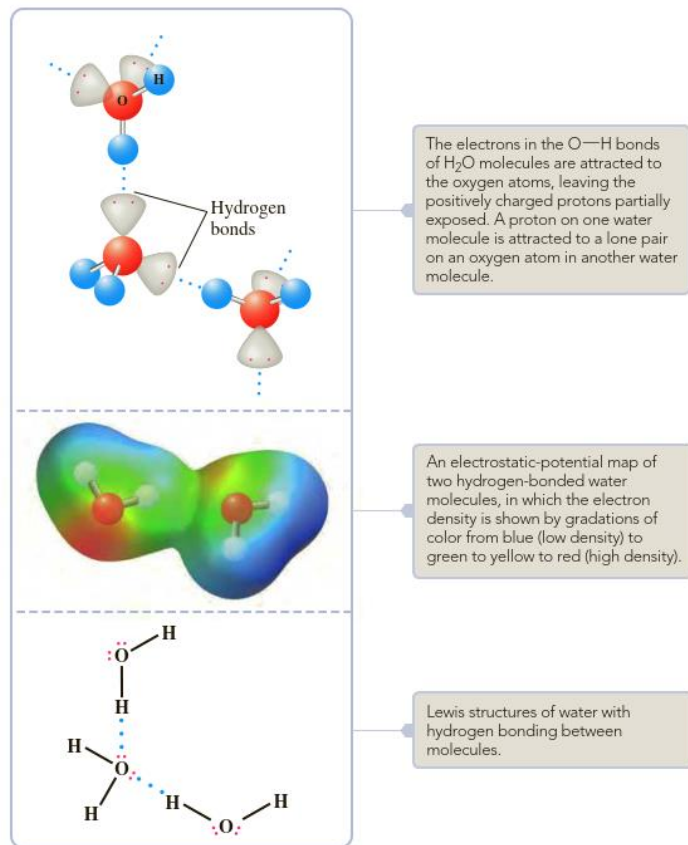
a Plot for the Group 6A hydrides. Note that the boiling points of H₂S, H₂Se, and H₂Te increase fairly smoothly, as you would expect if London forces are dominant. H₂O has a much higher boiling point, indicating an additional intermolecular attraction.



b Plot for the hydrides of Groups 7A, 5A, and 4A. Of the molecules on the plot, HF and NH₃ are the only ones that have hydrogen bonding.

Figure 11.24 ◀
Boiling point versus molecular weight for hydrides

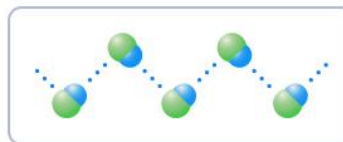
Figure 11.25 ▶
Hydrogen bonding in water



boiling points should increase regularly with molecular weight. You do see such a regular increase in boiling point for H_2S , H_2Se , and H_2Te . On the other hand, H_2O has a much higher boiling point than you would expect if only London forces were present. This is consistent with the view that hydrogen bonding exists in H_2O but is essentially absent in H_2S , H_2Se , and H_2Te .

Studies of the structure of H_2O in its different physical states show that hydrogen bonding is present. The hydrogen atom of one water molecule is attracted to the electron pair of another water molecule (Figure 11.25). Many H_2O molecules can be linked this way to form clusters of molecules. Similar hydrogen bonding is seen in other molecules containing —O—H groups.

Figure 11.24b shows plots of boiling point versus molecular weight for Groups 7A, 5A, and 4A hydrides. Hydrogen fluoride, HF, and ammonia, NH_3 , have particularly high boiling points compared with other hydrides of their periodic group. Definite structural evidence has established hydrogen bonding in HF. In solid hydrogen fluoride, the structure shows a zigzag arrangement involving hydrogen bonding.



Dipole–dipole forces

- › Attractions among molecules with dipole moments
- › Hydrogen bonding is a particularly strong form of dipole–dipole attraction
 - › Occurs in molecules containing hydrogen bonded to a highly electronegative element such as nitrogen, oxygen, or fluorine
 - › Produces unusually high boiling points

London dispersion forces

- › Caused by instantaneous dipoles that form in atoms or nonpolar molecules

- › Metallic bonding can be described in terms of two models
 - › Electron sea model: valence electrons circulate freely among the metal cations
 - › Band model: electrons are assumed to occupy molecular orbitals
 - › Conduction bands: closely spaced molecular orbitals with empty electron spaces

Metallic Bonding

In the early 1900's, Paul Drüde came up with the "sea of electrons" metallic bonding theory by modeling metals as a mixture of atomic cores (atomic cores = positive nuclei + inner shell of electrons) and valence electrons. Metallic bonds occur among metal atoms. Whereas ionic bonds join metals to non-metals, *metallic bonding joins a bulk of metal atoms*. A sheet of aluminum foil and a copper wire are both places where you can see metallic bonding in action.

Metals tend to have high melting points and boiling points suggesting strong bonds between the atoms. Even a soft metal like sodium (melting point 97.8°C) melts at a considerably higher temperature than the element (neon) which precedes it in the Periodic Table. Sodium has the electronic structure $1s^2 2s^2 2p^6 3s^1$. When sodium atoms come together, the electron in the 3s atomic orbital of one sodium atom shares space with the corresponding electron on a neighboring atom to form a molecular orbital - in much the same sort of way that a covalent bond is formed.

The electrons can move freely within these molecular orbitals, and so each electron becomes detached from its parent atom.

The electrons are said to be delocalized. The metal is held together by the strong forces of attraction between the positive nuclei and the delocalized electrons.

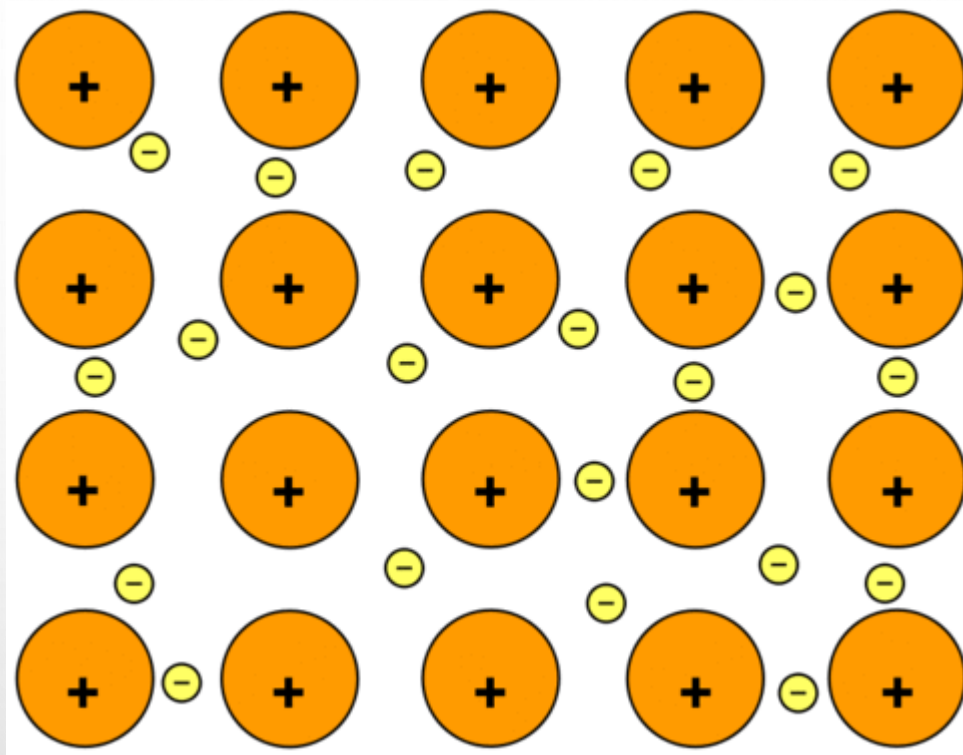


Figure: Metallic Bonding: The Electron Sea Model: Positive atomic nuclei (orange circles) surrounded by a sea of delocalized electrons (yellow circles).

Each positive center in the diagram represents all the rest of the atom apart from the outer electron, but that electron has not been lost - it may no longer have an attachment to a particular atom, but it's still there in the structure. Sodium metal is therefore written as Na, not Na^+ .